Model Cards and Human Review

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Model Cards and Human Review before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Proxy Features and Fairness

Without opening the old lesson, explain one key idea from Proxy Features and Fairness. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Improve Generalization

Without opening the old lesson, explain one key idea from Improve Generalization. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Overfitting

Without opening the old lesson, explain one key idea from Over**fitting**. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

Complete the supported Model Cards and Human Review trace and record the first step you can explain without help.

Use W2: Check arithmetic for **Model card**s and review. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Model Cards and Human Review explanation against the supplied facts before discussing it.

Use the worked case for **Model card**s and review. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply independently

Review this card: model v 2, 90 percent accurate, works for every school. No dataset size, split or baseline is given. Write a correction request with a bounded replacement claim.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

Connect the Model Cards and Human Review target to a trace and a justified conclusion.

Draft five model-card headings for a classroom leaf classifier. Why must the card state that its labels are classroom categories rather than professional plant diagnosis?

WORKBOOK · 6 OF 21

Solve the given case

Solve the supplied Model Cards and Human Review case and show the deciding evidence.

A **model card** says healthy/spotted,**80%** accuracy only.

Add responsible-use **evidence**.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

A **model card** says 90% accuracy but gives no **intended use**, data conditions or human-review rule. What should be added before classroom use?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 33 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Model Cards and Human Review adds to Responsible Plant Classifier and keep the evidence.

Complete the Plant Classifier **model card** and require teacher sign-off before any classroom demonstration.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Draft five model-card headings for a classroom leaf classifier. Why must the card state that its labels are classroom categories rather than professional plant diagnosis?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Draft a purpose

Draft a purpose. Apply the supplied Model Cards and Human Review facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Model Cards and Human Review", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Write an intended-use and non-use statement for the worked card.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Check arithmetic

> Check arithmetic. Apply the supplied Model Cards and Human Review facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Model Cards and Human Review", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Verify the worked accuracy, precision and recall.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Find missing evidence

> Find missing evidence. Apply the supplied Model Cards and Human Review facts and keep your reasoning.

A card says fair and reliable but provides only accuracy. Request three additions.

My reasoning and evidence

W4 Apply independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply independently. Apply the supplied Model Cards and Human Review facts and keep your reasoning.

Review this card: model v 2, 90 percent accurate, works for every school. No dataset size, split or baseline is given. Write a correction request with a bounded replacement claim.

My reasoning and evidence

W5 Maintain the record

> Maintain the record. Apply the supplied Model Cards and Human Review facts and keep your reasoning.

What should change in a card after a threshold revision?

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Model Cards and Human Review

> Write a model card whose performance statement can be traced to a concrete evaluation. Explain the labelled step before checking the changed condition.

Write a **model card** whose performance statement can be traced to a concrete evaluation.

Toy v = 3 classifier scored16/20 on five unseen plants with four views each, under indoor tablet lighting. Baseline scored14/20.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/33

Model Cards and Human Review

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Name evidence

Record v = 3, 20 photo cases, 5 plant identities, group-separated split and indoor tablet conditions.

2. Compare baseline

v = 3 accuracy 80%; baseline70% on the same 20 cases.

3. State limits

Do not claim all cameras, outdoor conditions or disease diagnosis were validated.

CHANGE ONE THING

The publicity sentence becomes "80% accurate for every plant".

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Write a bounded replacement for the publicity sentence. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Write a **model card** whose performance statement can be traced to a concrete evaluation.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Do not claim all cameras, outdoor conditions or disease diagnosis were validated.

1. Name **evidence**

2. State limits

3. Compare baseline

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: State limits Do not claim all cameras, outdoor conditions or disease diagnosis were validated. Read the picture in order.

Do not claim all cameras, outdoor conditions or disease diagnosis were validated. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The publicity sentence becomes “**80%** accurate for every plant”. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Reject that unbounded statement.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Reject that unbounded statement. The measured rate belongs to twenty related indoor photos, not every plant or setting.

Repair a missing building block

Review Proxy Features and Fairness (ai-4/32). Goal: Identify possible proxy features. Short practice: What **evidence** contradicts the original “all setting information removed” claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Improve Generalization (ai-4/30). Goal: Compare **training** and validation **evidence** for model capacity. Short practice: What fresh **evidence** is needed after adding crossed **training** data?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Over**fitting** (ai-4/27). Goal: Compare **training** and validation **evidence** for model capacity. Short practice: Does100% **training** accuracy prove Large was always best?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

WORKBOOK · 21 OF 21

Visual concept journey

Explain each Model Cards and Human Review picture from its labels, then record a relevant limit.

01. Model Cards and Human Review: the case

AI ACADEMY / AI-4/33 · 01 · LOOK AT THE FACTS

Model Cards and Human Review: the case

The situation

Write a model card whose performance statement can be traced to a concrete evaluation.

Supplied evidence

Toy v = 3 classifier scored16/20 on five unseen plants with four views each, under indoor tablet lighting. Baseline scored14/20. All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain model cards and human review using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

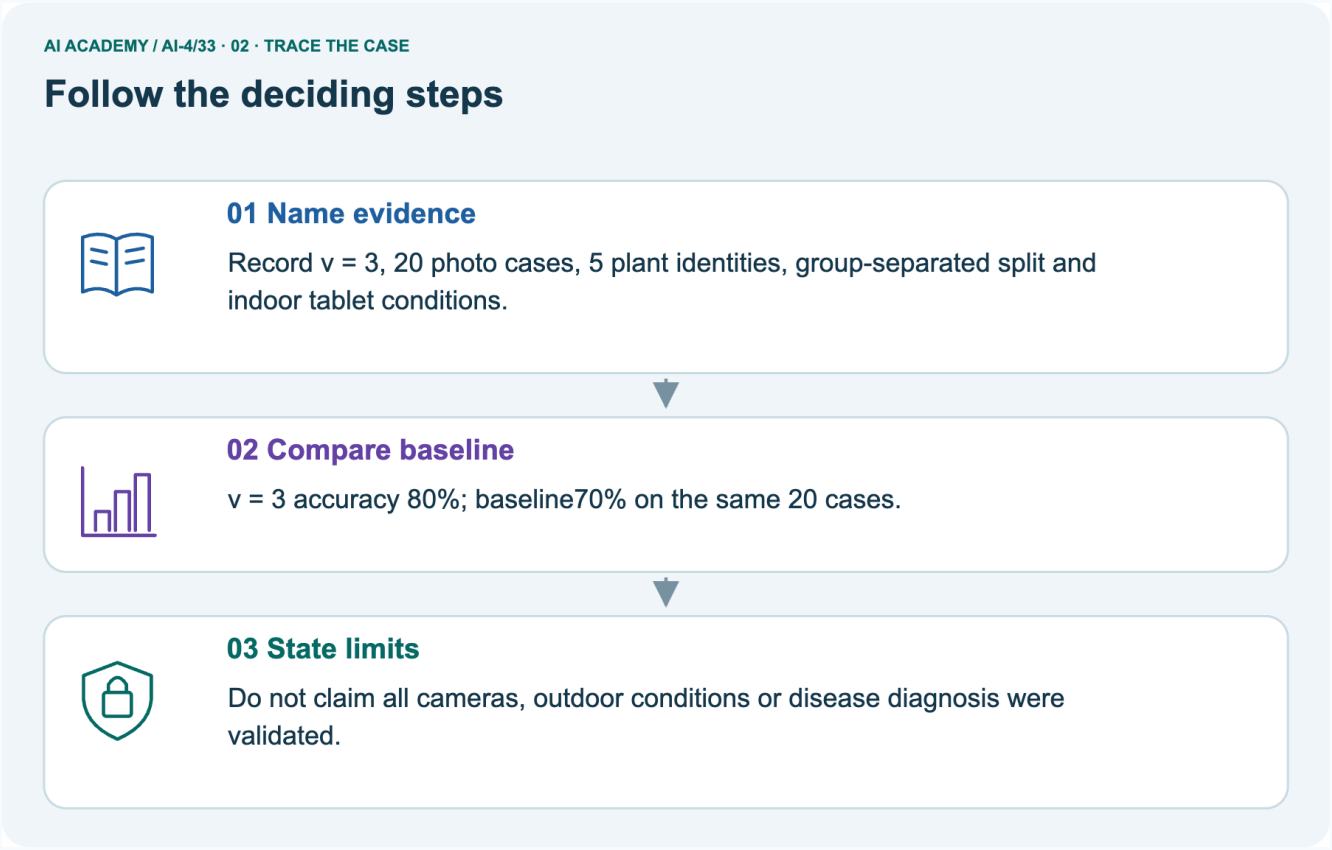
Write a **model card** whose performance statement can be traced to a concrete evaluation. Supplied facts: Toy v = 3 classifier scored16/20 on five unseen plants with four views each, under indoor tablet lighting.

Baseline scored14/20. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Name **evidence**: Record v = 3, 20 photo cases, 5 plant identities, group-separated split and indoor tablet conditions. → Compare baseline: v = 3 accuracy **80%**; baseline**70%** on the same 20 cases. → State limits: Do not claim all cameras, outdoor conditions or disease diagnosis were validated.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Model cards and review at a glance

AI ACADEMY / AI-4/33 · 03 · INSPECT THE LESSON MODEL

Model cards and review at a glance

Part	Evidence or action	Interpretation
Purpose and version	Toy classifier v 1	Fictional data only
Evidence	12 held-out cases	Full matrix and baseline
Limits and review	No real decisions	List unresolved coverage

The model card should name feature availability, preprocessing fitted on training, model settings, reference-label rules and test date. Include subgroup coverage if evaluated; otherwise say it was not assessed. Describe relevant failure examples without exposing personal information.

Follow the labelled picture with the matching explanation. Open the vector diagram

Model cards and review at a glance. Purpose and version | Toy classifier v 1 | Fictional data only. **Evidence** | 12 held-out cases | Full matrix and baseline. Limits and review | No real decisions | List unresolved coverage

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.

AI ACADEMY / AI-4/33 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

Do not claim all cameras, outdoor conditions or disease diagnosis were validated.

Changed condition

The publicity sentence becomes “80% accurate for every plant”.

Worked changed-case result

Reject that unbounded statement.
Why: The measured rate belongs to twenty related indoor photos, not every plant or setting.

Follow the labelled picture with the matching explanation. Open the vector diagram

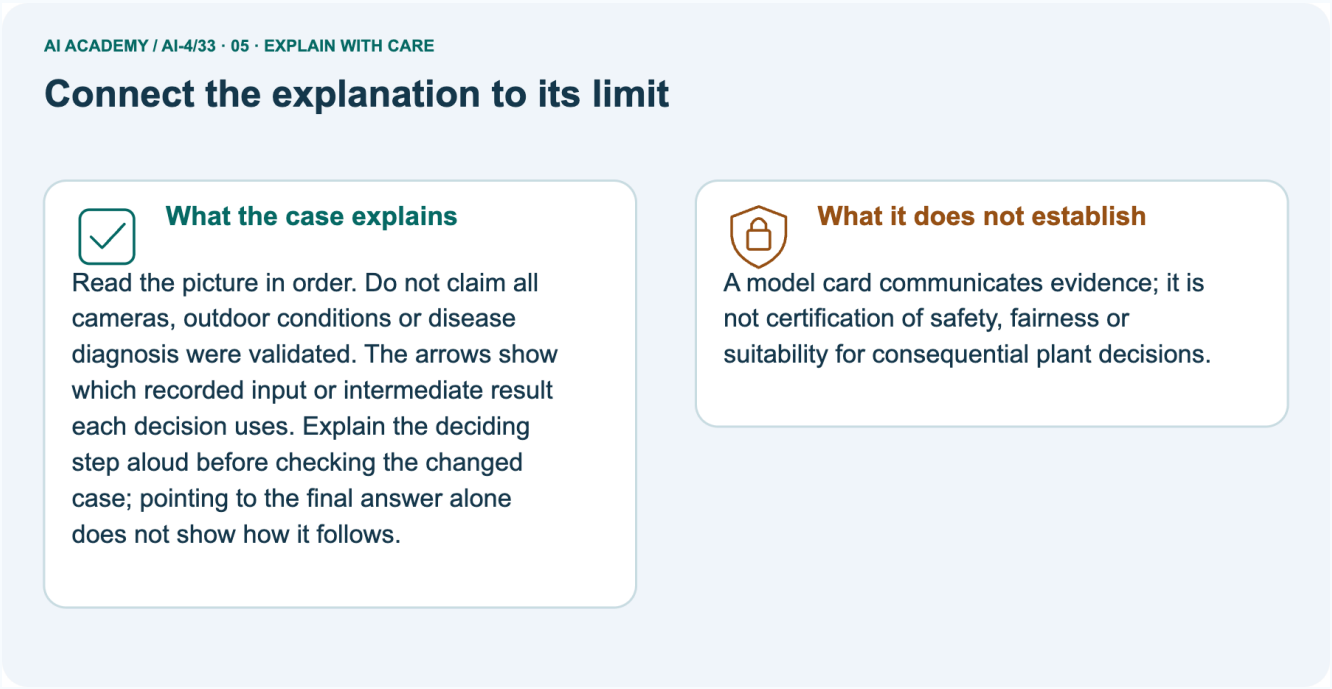
Original case: Do not claim all cameras, outdoor conditions or disease diagnosis were validated.
Changed condition: The publicity sentence becomes “**80%** accurate for every plant”.

Expected result: Reject that unbounded statement. Reason: The measured rate belongs to twenty related indoor photos, not every plant or setting.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Do not claim all cameras, outdoor conditions or disease diagnosis were validated. The arrows show which recorded **input** or intermediate result each decision uses.

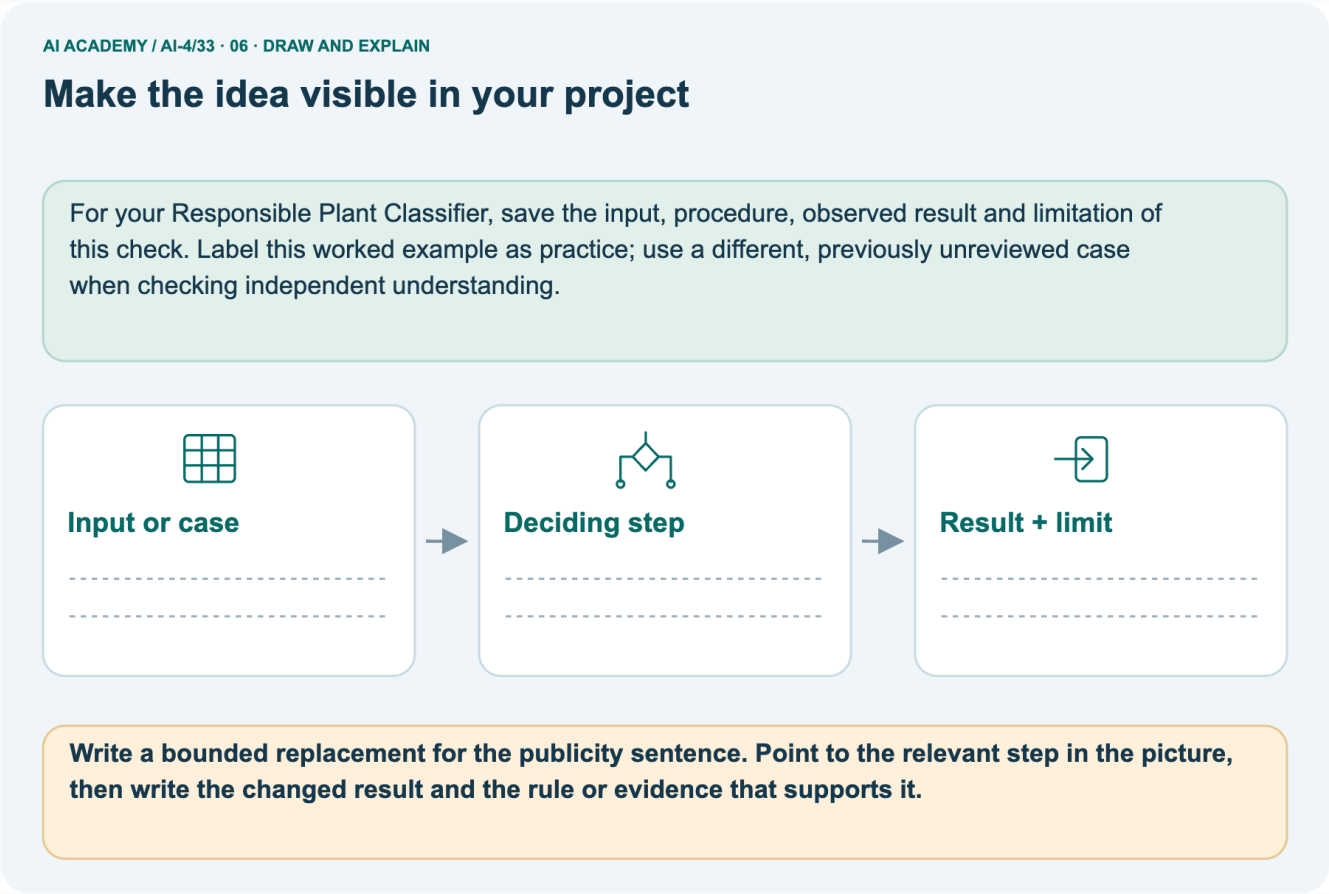
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A **model card** communicates **evidence**; it is not certification of safety, fairness or suitability for consequential plant decisions.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Write a bounded replacement for the publicity sentence. Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.